Week 12: November 16-20
Chemistry in Biology

Subject/:الموضوع Biology **Unit Title: عنوان الوحدة The Cell** **Gradeالصف : 9**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies Formative / Summative / Differentiation	Resources and links for Parents
Lesson 1	I can explain how plants capture light energy and convert it into chemical energy by identifying chloroplast structures, describing the role of pigments and photosystems, and sequencing the main events of the light-dependent reactions.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Photosynthesis.pdf
Lesson 2	I can explain how plants use ATP and NADPH to produce organic molecules by describing the Calvin cycle, interpreting its main stages, and analyzing how environmental conditions influence photosynthesis and plant adaptations.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Photosynthesis.pdf
Lesson 3	I can explain how cells release energy from glucose to produce ATP by describing glycolysis, the Krebs cycle, and electron transport, identifying their locations, inputs, outputs, and the roles of electron carriers.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Cellular Respiration.pdf
Lesson 4	I can compare aerobic respiration and fermentation by explaining how cells produce ATP under different oxygen conditions, modelling lactic-acid and alcohol fermentation, and evaluating the relationship between photosynthesis and cellular respiration.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Cellular Respiration.pdf
Homework **الواجبات**			

Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	

Week # 12 الأسبوع#

Technology Integration دمج التكنولوجيا	digital chloroplast model animated representation of the light-dependent reactions interactive Calvin cycle model digital mitochondrion model and animated pathway models
Cross-curricular links الروابط بين المواد الدراسية	Chemistry: Chemical equations, reactants and products, oxidation and reduction, and the role of electrons. Physics: Light energy, wavelengths, absorption, reflection, and energy conversion. Mathematics: Interpreting pigment absorption graphs and comparing wavelengths. Chemistry: Carbon fixation, molecular structures, and energy transfer through ATP and NADPH. Environmental Science: Plant adaptations to drought and heat. Chemistry: Oxidation and reduction, electron transfer, carbon compounds, and chemical equations. Physics: Energy transfer, chemical energy, and the relationship between energy changes and ATP production. Mathematics: Calculating net ATP production and organizing stage-by-stage data. Chemistry: Fermentation reactions, organic molecules, and oxidation–reduction. Food Technology: Yeast fermentation in bread-making and bacterial fermentation in yogurt and cheese production.
External Assessment Integration دمج التقييمات الخارجية	NA
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Research & Thinking (ATL): Interpret diagrams and explain how electron movement contributes to ATP and NADPH production. Communication: Use scientific vocabulary to describe the role of chloroplast structures and photosystems. Collaboration: Work with a partner to annotate the light-reaction diagram and explain the sequence. Research & Thinking (ATL): Interpret the Calvin cycle and evaluate how environmental factors affect photosynthesis. Communication: Explain plant adaptations using evidence and correct scientific vocabulary. Collaboration: Compare C4 and CAM pathways and justify plant selection for different environments. Research & Thinking (ATL): Analyze how the interruption of one stage affects the rest of cellular respiration.

	Communication: Explain energy transfer using accurate scientific vocabulary and labelled pathways.
Standards المعايير	HS-LS2-5: Develop a model to illustrate the role of photosynthesis and cellular respiration in the cycling of carbon among the biosphere, atmosphere, hydrosphere, and geosphere.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How do scientific models help us understand processes that occur at a scale too small to observe directly? How do scientists use indirect evidence to explain processes such as ATP production that cannot be observed directly in living cells?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 MUST DOK 1 – Recall: Identify the main chloroplast structures and state where the light-dependent reactions occur. SHOULD DOK 2/3 – Apply & Analyze: Explain how pigments, photosystems, water splitting, and electron transport contribute to ATP and NADPH production. COULD DOK 4 – Evaluate & Create: Construct and justify a labelled model of the light-dependent reactions and predict the effects of disrupting one stage. Lesson 2 MUST DOK 1 – Recall: Identify the location and main stages of the Calvin cycle and define carbon fixation. SHOULD DOK 2/3 – Apply & Analyze: Explain the roles of rubisco, ATP, and NADPH in G3P production and compare C4 and CAM adaptations. COULD DOK 4 – Evaluate & Create: Evaluate which plant adaptation is more suitable for a specified water-limited environment and justify the recommendation using biological evidence. Lesson 3 MUST DOK 1 – Recall: Identify the stages and locations of cellular respiration and define ATP, NADH, and $FADH_2$. SHOULD DOK 2/3 – Apply & Analyze: Explain how glucose is broken down and how electron carriers contribute to ATP production during aerobic respiration. COULD DOK 4 – Evaluate & Create: Construct and justify a cause-and-effect model predicting how interruption of oxygen availability affects the stages of aerobic respiration. Lesson 4 MUST DOK 1 – Recall: Identify lactic-acid and alcohol fermentation and state their products.

	SHOULD DOK 2/3 – Apply & Analyze: Explain how fermentation regenerates NAD^+ and compare aerobic respiration with fermentation. COULD DOK 4 – Evaluate & Create: Evaluate the claim that plants do not need cellular respiration and construct an evidence-based model showing the relationship between photosynthesis and cellular respiration.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Citizenship — Conservation Values – Compassion
	Area: Environmental conservation, sustainable agriculture, and protection of plant life in UAE ecosystems. Area: Public health, physical wellbeing, and understanding the biological basis of energy requirements during daily activities.
	How to Implement: Discuss the importance of photosynthetic organisms in maintaining oxygen levels and supporting food webs. Connect plant productivity to UAE environmental initiatives, including the protection of native vegetation and mangroves. Students explain why conserving photosynthetic organisms supports ecosystem stability. How to Implement: Connect cellular respiration to the energy requirements of the human body during exercise, work, and daily activities. Students discuss how understanding the importance of oxygen and ATP can support informed decisions about physical activity, health, and wellbeing within the UAE community.